

Technology Stack

Field	Details
Date	05-07-2026
Team ID	(Mee Team ID)
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Define Your Technology Stack

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend	HTML	Used to create the structure of the web application.
2	Frontend	CSS	Used to design a responsive and user-friendly interface.
3	Backend	Flask	Handles user requests, routing, and communication between the frontend and ML model.
4	Programming Language	Python	Used to develop the machine learning model and application logic.
5	Data Processing	Pandas	Used to load, clean, and preprocess the crop recommendation dataset.
6	Numerical Computing	NumPy	Used for numerical operations and preparing input data for prediction.
7	Machine Learning	Scikit-learn	Used to train the crop recommendation model and make predictions.
8	Model Storage	Joblib	Used to save and load the trained machine learning model (crop_model.pkl).
9	Dataset	Crop Recommendation Dataset	Contains soil and environmental parameters used for training the model.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
10	Version Control	Git & GitHub	Used for source code management and project collaboration.
11	Development Environment	Visual Studio Code	Used for coding, testing, and debugging the application.